

Vortex-induced vibration control of bridge decks using energy dissipative devices: A review of recent developments

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ABSTRACT

This paper presents a comprehensive and up-to-date review of energy dissipative devices for controlling vortex-induced vibrations (VIV) in bridge decks, covering passive, active, and semi-active control systems. It synthesizes key findings and innovative control strategies from the VIV control research community, critically evaluating the advantages and limitations of each device type and highlighting essential research gaps for future exploration. A systematic literature review identifies and summarizes twelve documented real-bridge applications of these devices: ten involving tuned mass dampers (TMDs), one with viscous dampers, and one employing semi-active TMDs. While TMDs have demonstrated effectiveness, ongoing research focuses on optimizing their parameters under nonlinear vortex-induced forces, closely spaced modal frequencies, and multi-objective criteria. Emerging research trends emphasize the development of advanced passive devices for ultra-low-frequency VIV control, lightweight and compact designs, and multimodal vibration mitigation. Furthermore, active and semi-active control devices show considerable promise due to their superior adaptability and enhanced performance over passive systems, indicating a promising direction for future VIV mitigation strategies in long-span bridges.

1. Introduction

Long-span bridges are particularly vulnerable to vortex-induced vibrations (VIVs) during both construction and operational stages. Although VIV rarely causes immediate structural damage or failure, it can significantly degrade driving comfort and safety, potentially resulting in traffic interruptions and fatigue damage to critical components (Zhao et al., 2022). These effects may lead to substantial economic losses and adverse social consequences. In real-world scenarios, VIV predominantly appears as vertical oscillations, while torsional VIV is relatively uncommon. Table 1 summarizes documented cases of long-span bridges that have experienced pronounced vertical VIV events.

Given the frequent occurrence and potential consequences of VIVs in long-span bridges, their control has become a critical focus in both bridge engineering and wind engineering (Fujino and Siringoringo, 2013; Gao et al., 2021). Control measures for bridge VIV are primarily divided into aerodynamic and mechanical approaches. Aerodynamic measures, often considered the first line of defense, seek to modify the bridge girder's aerodynamic profile by installing or optimizing

components such as railings, fairings, guide vanes, vortex suppressors, stabilizers, and maintenance rails to reduce or mitigate VIV in bridges (Sarwar and Ishihara, 2010; Zhao et al., 2019; Wang et al., 2023a). During the design phase of a long-span bridge, systematic aerodynamic optimization of the girder's geometry based on wind tunnel tests is routinely carried out to reduce VIV risk. However, VIV behavior is sensitive to factors including girder geometry, auxiliary attachments, structural damping ratios, and Reynolds number effects (Wu and Kareem, 2012; Li et al., 2014; Wang et al., 2024a). Aerodynamic control measures are typically designed and validated using small-scale wind tunnel tests. Due to inherent experimental uncertainties and scale effects, these tests sometimes struggle to accurately replicate the VIV response of full-scale bridges. As a result, some aerodynamic measures that prove effective in wind tunnel experiments may underperform or fail to achieve the expected benefits in actual bridge applications.

When the application of aerodynamic measures is impractical or proves insufficient, mechanical measures, typically involving the installation of energy-dissipative devices, are often employed to control VIV. Tuned mass dampers (TMDs) are the most widely used energy

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Table 1
Bridges that experienced VIVs.

Bridge Type	Bridge	Location	Span arrangement (m)	Maximum amplitude (cm)	Reference
Girder bridge	Great Belt East approach Bridge	Denmark	140 + 2 × 193 + 92	10.0	Larsena et al. (1995)
	Rio-Niterói Bridge	Brazil	74 + 200 + 300 + 200 + 74	25.0	Battista and Pfeil (2010)
	Trans-Tokyo Bay Crossing Bridge	Japan	... + 190 + 2 × 240 + 190 + ...	54.0	Fujino and Yoshida (2002)
	Volgograd Bridge	Russia	... + 3 × 155 + ...	70.0	Corriols and Morgenthal (2012)
Suspension bridge	Chongqi Bridge	China	102 + 4 × 185 + 102	30.0	Sun et al. (2020)
	Original Tacoma Bridge	USA	335 + 853 + 335	76.0	Matsumoto et al. (2003)
	Deer Isle Bridge	USA	61 + 329 + 61	5.5	Kumarasena et al. (1991)
	Great Belt East Bridge	Denmark	535 + 1624 + 535	50.0	Larsen et al. (2000)
	Osterøy Bridge	Norway	595 (main span)	50.0	Isaksen et al. (2001)
	Xihoumen Bridge	China	578 + 1650 + 585	24.0	Li et al. (2014)
	Yi Sun-sin Bridge	South Korea	357.5 + 1545 + 357.5	26.0	Kim et al. (2016)
	Humen Bridge	China	302 + 888 + 348.5	30.0	Ge et al. (2022)
	Yingwuzhou Bridge	China	200 + 2 × 850 + 200	50.0	Cao et al. (2023)
	Verrazzano Bridge	USA	370 + 1300 + 370	un	un
	Çanakkale 1915 Bridge	Turkey	680 + 2023 + 265	un	un
Cable-stayed bridge	Wye Bridge	Canada	235 (main span)	3.5	Smith (1980)
	Ishikari Kako Bridge	Japan	64 + 160 + 64	10.0	Burden et al. (1991)
	Kessock Bridge	UK	80 + 240 + 80	11.0	Owen et al. (1996)
	Fred Hartman Bridge	USA	147 + 380 + 147	50.0	Ozkan et al. (2001)
	Second Severn Crossing Bridge	UK	120 + 456 + 120	20.0	Macdonald et al. (2002)
	Second Jindo Bridge	South Korea	70 + 344 + 70	12.0	Seo et al. (2013)

un: unavailable from published literature.

Table 2
Bridges utilizing energy dissipative devices for VIV control.

Device	Bridge	Vertical VIV modes	Number of devices	Mass per device (t)	Reference
Passive TMD	Kansai Airport access Bridge	un	un	un	Honda et al. (1993)
	Great Belt West approach Bridge	1st and 2nd	4	un	Larsena et al. (1995)
	Great Belt East approach Bridge	1st to 4th	12	un	Larsena et al. (1995)
	Kessock Bridge	1st	8	2	Owen et al. (1996)
	Trans-Tokyo Bay Crossing Bridge	1st and 2nd	16 (8 for each mode)	10	Fujino and Yoshida (2002)
	Rio-Niterói Bridge	1st	32	2.2	Battista and Pfeil (2010)
	Chongqi Bridge	1st	16	3.6	Meinhardt et al. (2014)
	Second Jindo Bridge	1st	5	0.61	Ayoub (2015)
	Hong Kong-Zhuhai-Macao Bridge	1st	92	4 or 4.5	Dai et al. (2019)
	Humen Bridge	3rd, 4th, and 5th	un	un	un
Semi-active TMD	Volgograd Bridge	1st to 3rd	12 (4 for each mode)	4	Weber et al. (2013)
Viscous damper	Yingwuzhou Bridge	7th	4	0.6	Cao et al. (2023)

Note: Information regarding the TMDs of the Humen Bridge is based on private communication with the TMD designer.

un: unavailable from published literature

dissipative devices for mitigating VIV. As summarized in Table 2, a lot of bridges have successfully implemented TMDs for VIV control (Ayoub, 2015; Battista and Pfeil, 2010; Dai et al., 2019; Honda et al., 1993; Larsena et al., 1995; Meinhardt et al., 2014; Owen et al., 1996; Fujino and Yoshida, 2002).

Although TMDs have proven to be effective for controlling VIVs in bridges, they also have certain limitations. For instance, as the controlled modal frequency decreases, the static deformation of the TMD spring, due to the combined weight of the mass block and the spring itself, increases rapidly, posing design and installation challenges (Xu et al., 2019). Additionally, conventional TMDs are typically effective within a narrow frequency band and are tuned to a single vibration mode. In scenarios involving multimode VIVs, multiple TMD units must be installed, which increases system complexity and cost (Huang et al., 2025). To address these challenges, various improved forms of TMDs have been developed in recent years, targeting enhanced control performance, robustness, and structural integration.

Active and semi-active control devices are promising alternatives that offer adaptive capabilities and enhanced vibration mitigation performance (Casciati et al., 2012; Soong and Spencer, 2002; Spencer

and Nagarajaiah, 2003). Active devices employ external energy sources and real-time feedback control to dynamically adjust control forces, achieving superior performance in complex and time-varying environments. Meanwhile, semi-active devices, such as magnetorheological (MR) dampers, provide a compromise solution that combines the adaptability of active systems with the reliability and low power consumption of passive devices.

This paper provides a comprehensive review of recent research and developments in the application of energy dissipative devices for VIV control in bridges. It also provides a detailed overview of the practical implementations of these devices in existing bridges. It should be noted that the purpose of this review is not to exhaustively cover all types of energy dissipative devices or their developments across the broad field of vibration control, but rather to concentrate specifically on their research progress and applications within the field of VIV control in bridge decks. Aerodynamic countermeasures, as well as VIV control for other structural components such as bridge towers and cables, are beyond the scope of this review. Recent reviews covering TMDs and other energy dissipative devices include works by Ma et al. (2021), Liu et al. (2022), Ding and Chen (2020), and Li et al. (2025); however, none of

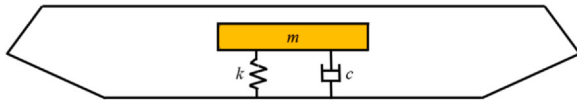


Fig. 1. Schematic diagram of a TMD installed in a bridge deck.

these specifically focus on VIV control of bridge decks.

The remainder of this paper is organized as follows. Section 2 presents an overview of practical applications of TMDs in long-span bridges for VIV control, along with recent advancements in TMD optimization methods and the development of TMD variants for low-frequency and lightweight applications. This section also discusses other passive devices, including nonlinear energy sinks (NESs) and viscous dampers. Sections 3 and 4 review recent investigations into active and semi-active control strategies for bridge VIV control, respectively. Section 5 provides recommendations for future research directions. Finally, Section 6 concludes the review.

2. Passive device

2.1. Tuned mass damper (TMD)

The TMD is one of the most widely adopted energy dissipative devices for VIV control of bridge decks. A typical TMD consists of an auxiliary mass block, a stiffness element (commonly in the form of coil springs), and a damping element (typically a viscous damper). A schematic diagram is illustrated in Fig. 1, where m , k , and c represent the effective mass, stiffness coefficient, and damping coefficient of the TMD, respectively. The schematic diagram in Fig. 1 employs compression springs as the stiffness component; however, it is worth noting that many TMDs also use extension springs in practice. By tuning the natural frequency of the auxiliary mass to match that of the primary structure, the vibrational energy can be effectively transferred to the mass block and dissipated through the damping element. This process significantly reduces the vibration amplitude of the primary structure.

In bridge engineering applications, the damping element of a TMD is typically realized using a viscous damper. While effective, viscous dampers may suffer from issues such as fluid leakage, temperature sensitivity, and long-term maintenance requirements. To address these limitations, eddy-current dampers have been proposed as an alternative, giving rise to the eddy-current TMD (Wang et al., 2012). By leveraging magnetic induction to generate damping forces, this configuration eliminates mechanical contact and fluid components, thereby avoiding wear and leakage. As a result, eddy-current TMDs offer high durability, stable performance over a wide temperature range, and minimal maintenance needs, which are particularly advantageous for long-span bridges where inspection and servicing are challenging. Alternatively, the damping element can be replaced or supplemented with a pounding mechanism, giving rise to the pounding TMD (Song et al., 2016). This enhanced TMD incorporates nonlinear impact interactions that expand the operational bandwidth and improve control effectiveness under large or variable excitations. Notable investigations into the application of eddy-current TMDs for bridge VIV control include the works of Huang (2016) and Chen et al. (2025), whereas research on pounding TMDs is represented by Wang et al. (2018).

2.1.1. Implementation of TMDs in VIV control of real bridges

Several practical implementations of TMDs for VIV control in real bridge projects have been reported in the literature. Representative case studies are summarized below and in Table 2.

Honda et al. (1993) reported the installation of 26 TMDs to control the VIVs of the Kansai International Airport access Bridge in Japan. Field measurements indicated that the damping ratio of the bridge increased from 0.02–0.03 to approximately 0.10 after implementation.

Larsena et al. (1995) described the design of TMDs for the Great Belt West and East Approach Bridges. The TMDs were intended to suppress

VIVs occurring at wind speeds up to 25 m/s. For the West Approach Bridge, two vibration modes were targeted, while four modes were addressed in the East Approach Bridge.

The Kessock Bridge, a cable-stayed bridge with a main span of 240 m completed in 1982, was equipped with 8 TMDs, each weighing 2 t, to control VIVs in its first vertical mode (0.51 Hz). Owen et al. (1996) reported significant performance deterioration of the TMDs over time, including loss of damping fluid, physical damage, and piston misalignment. These issues were speculated to stem from an insufficient mass travel allowance (designed as ± 150 mm), highlighting the importance of robust and long-term TMD design for bridge applications.

The Trans-Tokyo Bay Highway Crossing Bridge, completed in 1997, is a ten-span continuous steel box-girder bridge with a total length of 1630 m. The two longest spans each measure 240 m. Significant VIVs were observed when wind speeds ranged between 13 m/s and 18 m/s, with a maximum vibration amplitude of approximately 54 cm. Fujino and Yoshida (2002) reported the implementation of a TMD system designed to control the first two vibration modes of the bridge, with natural frequencies of 0.33 Hz and 0.47 Hz. The system consists of 16 TMDs (8 tuned to each mode), with each TMD having a mass of 10 t.

The Rio-Niterói Bridge is a continuous beam bridge completed in 1974. The three central spans (200 + 300 + 200) m of the bridge consist of continuous steel twin box girders. Over the years, the bridge has experienced VIVs when wind speeds reached approximately 15–16.5 m/s. These vibrations were associated with the first vertical mode, which has a natural frequency of 0.32 Hz, and exhibited maximum amplitudes of up to 0.60 m at midspan. Battista and Pfeil (2010) reported the implementation of a TMD system comprising 32 TMDs, each with a mass of 2.2 t.

The Chongqi Bridge is a six-span continuous beam bridge with a span arrangement of (102 + 3 × 185 + 102) m. The bridge was opened to traffic in 2011. During its construction, VIVs were observed when the wind velocity reached approximately 15 m/s. To mitigate these vibrations, a multiple TMD (MTMD) system was designed and installed, targeting the first vibration mode with a natural frequency of 0.52 Hz. A total of 16 TMDs were deployed, each with a mass of 3.6 t (Meinhardt et al., 2014; Sun et al., 2020).

The Second Jindo Bridge is a cable-stayed bridge completed in 2005, featuring a main span of 340 m. In April 2011, the bridge experienced VIVs, during which the acceleration exceeded 1.5 m/s². To address this issue, an MTMD system was implemented. Ayoub (2015) reported that the system consists of 5 TMDs, each with a mass of 0.61 t, and with natural frequencies distributed between 0.425 Hz and 0.449 Hz.

Dai et al. (2019) reported the TMD system designed for the Hong Kong-Zhuhai-Macao Bridge, targeting the first vertical mode with a frequency of 0.88 Hz. The system consists of 92 TMDs, each with a mass of 4 or 4.5 t.

The Humen Bridge is a long-span suspension bridge with a main span of 888 m. It was opened to the public in 1997 and had been in operation for nearly 23 years before experiencing VIVs in 2020. The VIVs were triggered by the temporary installation of water-filled barriers and occurred in three vertical modes, with natural frequencies of 0.230, 0.276, and 0.367 Hz (Ge et al., 2022; Wang et al., 2023b). Although the vibrations were successfully mitigated through two aerodynamic measures, namely the addition of flow-suppressing plates and the removal of the outer maintenance rails (Ge et al., 2022), a TMD system was also installed as a secondary safety measure.

2.1.2. Recent development on TMD optimization

Although TMD technology is relatively mature and widely implemented in engineering practice, the optimization of TMD parameters remains an active area of research. In typical applications for VIV control, the TMD is designed based on a simplified two-degree-of-freedom model comprising the target vibration mode and the TMD itself (Strommen and Hjorth-Hansen, 2001; Yu et al., 2023). The device is usually positioned at the location of maximum modal displacement for

the target mode, and its mass is determined according to control objectives, economic considerations, and other practical constraints. Subsequently, the stiffness and damping parameters of the TMD are optimized for the assumed two-degree-of-freedom system. Traditionally, this optimization process either neglects the influence of vortex-induced forces or incorporates them using simplified models with limited fidelity.

Dai et al. (2019) compared TMD optimization outcomes using Scanlan's nonlinear vortex-induced force model and an empirical linear model. For a fixed mass ratio, both models yielded comparable optimal frequency ratios; however, the optimal damping ratio derived from Scanlan's model was significantly lower. In a follow-up study, Dai et al. (2020) evaluated the performance of TMDs designed using three theoretical formulations: free vibration, harmonically forced vibration, and self-excited vibration-based formulations. The results indicated that the TMD designed based on the self-excited vibration formulation achieved the most effective vibration mitigation, while the design based on the forced vibration formulation exhibited the greatest robustness to variations in the frequency ratio. Yu et al. (2023) further examined the sensitivity of TMD optimization to different vortex-induced force models. Their findings indicated that both the empirical linear model and Scanlan's nonlinear model tend to overestimate the effectiveness and robustness of TMDs. As more accurate alternatives, they recommended the use of a generalized polynomial model or an aerodynamic envelope model in the optimization of TMDs for VIV control (Zhang et al., 2020).

To enhance control robustness, MTMDs with slightly different natural frequencies distributed around the target modal frequency are often employed. In the context of bridge VIV control, the parameters of the MTMD system on the Second Jindo Bridge were determined following the optimization method proposed by Abé and Fujino (1994), as reported by Ayoub (2015). Peng et al. (2022) proposed a derivative-based optimization approach to determine the optimal parameters of MTMDs. With the same total mass ratio, MTMDs demonstrated superior performance compared to single TMD. Specifically, while the single TMD scheme typically results in a sharp peak-valley shape in the frequency response curve, the MTMD scheme yields a flatter plateau over a broader frequency range, indicating improved robustness in vibration suppression.

The VIVs of a bridge may be governed by different structural modes as wind velocity varies. In such scenarios, TMDs should be designed for each relevant mode. Previous studies have predominantly adopted a mode-by-mode design strategy for multi-mode VIV control, in which TMDs are independently designed for each structural mode. In this approach, the design of a TMD for a given mode typically considers only the target mode, while the possible interactions with other modes are disregarded. Consequently, the TMD parameter optimization methods discussed earlier remain applicable. For bridge structures with well-separated modal frequencies, this mode-by-mode design approach generally achieves satisfactory vibration mitigation performance.

However, for bridges with closely spaced modal frequencies, the conventional mode-by-mode TMD design method may exhibit significant limitations (Xu et al., 2023a; Huang et al., 2025). The primary concern lies in the fact that the control force generated by a TMD designed for the target VIV mode acts as an external load on the bridge, which may in turn excite forced vibrations in non-resonant modes. This results in an undesirable transfer of vibration energy from the target VIV mode to adjacent non-resonant modes. As a consequence, although the response of the target mode may be effectively suppressed, the response of other modes, particularly those with close frequencies, can increase significantly. This energy redistribution effect leads to a discrepancy between the theoretical control performance predicted under the single-mode assumption and the actual control effect. Therefore, TMDs designed using the mode-by-mode strategy may fail to achieve the desired suppression effect in such cases. For example, in the aeroelastic wind tunnel tests of the Great Belt East approach Bridge, it was

observed that TMDs designed for the first two modes interfered with each other, significantly reducing the overall control performance (Larose et al., 1995).

To address the limitations of the conventional mode-by-mode TMD design approach, several researchers have explored the use of advanced optimization algorithms to simultaneously optimize TMD parameters and layout, particularly in the presence of modal coupling effects. Modal coupling refers to the interaction between multiple vibration modes that coexist in the dynamic response of a bridge-TMD system, particularly when the bridge exhibits closely spaced natural frequencies (Huang et al., 2025). Xu et al. (2022b) applied a genetic algorithm to design six inerter-based TMDs for controlling the VIVs of the first six modes of a cable-stayed bridge and achieved satisfactory control performance. Xu et al. (2023a) used a particle swarm optimization algorithm to optimize the design of TMDs for controlling the first six modes of a six-span continuous beam bridge, simultaneously tuning 24 parameters including the location, mass, damping ratio, and frequency ratio of six TMDs. However, these optimization algorithms often involve high computational costs and are prone to converging to local optima. This issue becomes especially critical in multi-mode VIV control problems, where the number of design parameters is large and the complexity of the optimization problem increases significantly. Consequently, simplifying assumptions are often introduced to reduce the computational burden. Huang et al. (2025) found that by properly optimizing the number and spatial distribution of distributed TMDs, the modal coupling effect induced by TMDs can be significantly reduced or even completely eliminated. Building on this insight, they proposed a design principle for deploying distributed TMDs aimed at minimizing or eliminating modal coupling, and further developed an optimization methodology tailored for stable and user-friendly multi-mode VIV control in bridges with closely spaced modal frequencies.

Previous studies have primarily focused on optimizing TMD parameters to reduce the VIV amplitude. However, a recent trend is to adopt multi-objective optimization approaches that consider multiple performance criteria simultaneously. For example, Patil et al. (2011) proposed a multi-objective optimization method to mitigate VIV in long-span bridges by jointly optimizing cross-sectional retrofits and TMDs, achieving a balance between performance enhancement and cost reduction through Pareto-optimal solutions. Dai et al. (2022b) developed a parameter design framework for MTMD systems based on failure probability, which serves as the objective function by accounting for uncertainties in bridge properties, aerodynamic effects, and MTMD parameters. Their results highlighted that uncertainties in the natural frequencies of both the controlled bridge modes and the MTMDs significantly influence the equivalent damping ratio provided by the MTMD system. Xu et al. (2023b) further advanced this field by proposing a multi-objective optimization approach for mass dampers that balances control effectiveness and system stability in bridge VIV mitigation, enabling more robust and efficient damper designs.

2.2. TMD variants for low-frequency VIV control

A major limitation of the traditional TMD is the substantial static deformation experienced by the spring as the controlled modal frequency decreases. This deformation, resulting from the combined weight of the TMD mass block and the spring itself, increases rapidly at lower frequencies. The static deformation can be expressed as:

$$\Delta = mg/k = g/(2\pi f)^2 \quad (1)$$

where g is the gravitational acceleration, and f is the natural frequency of the TMD, which is approximately equal to the controlled modal frequency.

To mitigate the excessive static deformation of TMD springs at low frequencies, several improved TMD variants have been developed. One notable variant is the lever-type TMD. For instance, the Tokyo Bay Bridge in Japan utilized two sets of lever-type TMDs with natural

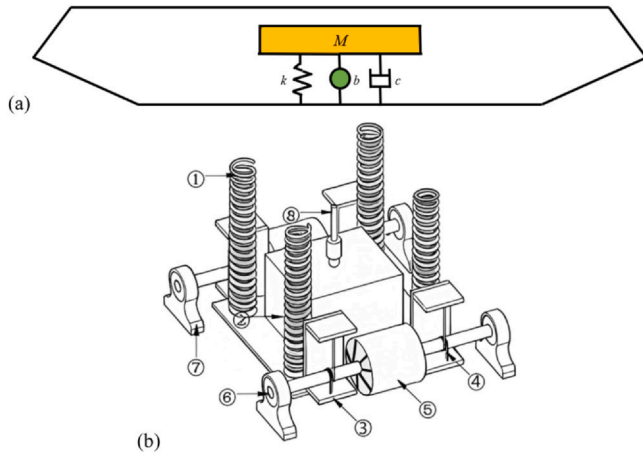


Fig. 2. Schematic diagrams of (a) a TMDI installed in a bridge deck and (b) a TMDI device proposed by the present authors (Wang et al., 2025).

frequencies of 0.33 Hz and 0.47 Hz, respectively. By employing a lever mechanism with a 5:1 ratio, the static deformation of the spring is reduced to one-fifth of that in a conventional TMD (Fujino and Yoshida, 2002). Although the lever-type TMD effectively mitigates spring deformation, it requires a complex supporting structure, which substantially increases the overall system weight and introduces complications such as additional joints and localized structural reinforcement.

An alternative approach was proposed by Wang et al. (2024b), who developed an innovative tuned liquid mass damper. In this design, the spring-supported mass block is submerged in a water tank, leveraging the added mass effect of the fluid to achieve a dynamic mass equivalent to that of a conventional TMD while employing a lighter physical mass. Additionally, the buoyancy provided by the water reduces the vertical load on the spring, thereby significantly decreasing the static deformation. A prototype of the tuned liquid mass damper was fabricated, featuring an equivalent mass of approximately 2 t and a natural frequency of 0.191 Hz. The measured static deformation of the spring was only 2.171 m, substantially lower than 6.805 m observed in a conventional TMD. The damper's vibration control performance was validated through both laboratory experiments and field tests on a cable-stayed bridge, where the frequency of the 1st-order vertical mode is 0.322 Hz.

Another promising enhancement involves the incorporation of inerter elements into TMDs, as shown in Fig. 2(a). An inerter is a two-terminal mechanical device capable of producing an apparent mass significantly larger than its physical mass by converting linear motion into high-speed rotational motion. For an ideal linear inerter, the force generated is proportional to the relative acceleration between its two terminals. Originally developed for suspension systems in Formula 1 racing, inerters have since found applications in civil engineering, particularly in seismic isolation and structural vibration control. Relevant progress and applications are comprehensively reviewed in recent literature (Smith, 2020; Ma et al., 2021; Wagg, 2021; Liu et al., 2022; Jha et al., 2025). A TMD equipped with an inerter, commonly referred to as a tuned mass damper inerter (TMDI), integrates the inerter in parallel with the damping and stiffness elements. The natural frequency of a TMDI is given by:

$$f = \frac{1}{2\pi} \sqrt{k/(m+b)} \quad (2)$$

where b is the inertance of the inerter. Compared to a traditional TMD, the spring stiffness k required to achieve the same natural frequency in a TMDI is significantly higher. This results in reduced static deformation and enhances the feasibility of employing TMDIs for low-frequency VIV control in long-span bridges.

A number of studies have explored the application of TMDIs for low-frequency VIV mitigation of bridges. Xu et al. (2019) and Dai et al.

(2019) were among the first to numerically examine the use of TMDIs for controlling VIV in bridge structures. Their results demonstrated that the TMDI configuration exhibits significantly reduced static deformation compared to conventional TMDs. Xu et al. (2020) conducted a numerical comparison of the VIV control performance of four inerter-based dampers, each comprising distinct combinations of dashpot and spring elements arranged either in series or in parallel with the inerter. Based on these simulations, a design guideline for VIV-resistant long-span bridges employing inerter-based dampers was proposed. Dai et al. (2021a) examined a novel TMDI configuration in which the conventional dashpot is replaced by a Maxwell element, showing promising VIV control effectiveness. Xu et al. (2022a) proposed closed-form design formulas for TMDIs specifically tailored for bridge VIV suppression. In a subsequent study, Xu et al. (2022b) further developed a strategy for multi-mode VIV mitigation in long-span cable-stayed bridges using spatially distributed TMDIs. Dai et al. (2024) proposed an approach to improve control robustness under system uncertainties by deploying multiple TMDIs with non-identical design parameters. More recently, Han et al. (2025) proposed integrating a pulley block into the TMDI system to redirect the relative displacement between the bridge and the mass block into the longitudinal direction of the bridge. This configuration allows control elements such as dampers, springs, and inerters to be installed along the bridge axis. Consequently, the spring in the pulley-block TMDI can be positioned outside the TMDI assembly, eliminating constraints imposed by the limited height of the box girder.

Previous numerical studies typically assumed the inerter to be an ideal linear element, wherein the force generated is directly proportional to the relative acceleration between its two terminals, without accounting for potential friction or nonlinearities. However, practical inerters inevitably exhibit nonlinear behavior and frictional effects that can influence their actual dynamic characteristics. Li et al. (2022) investigated the nonlinearity of a ball-screw inerter and its impact on the control effectiveness of TMDIs for suppressing bridge VIVs. Their results indicated that the dynamic properties of the ball-screw inerter depend on the magnitude of the input acceleration, and that the control performance of a TMDI incorporating a nonlinear inerter is inferior to that of an ideal linear TMDI. Peng et al. (2025a) experimentally evaluated a prototype TMDI incorporating a rack-and-pinion inerter. Their observations revealed significant spiking characteristics in the acceleration response of the TMDI, which may be caused by the opening and closing of the backlash between the gear and rack. Unlike the ideal TMDI, Coulomb friction within the mechanical components had a notable impact on the system's dynamic response, manifesting in amplitude-dependent behavior and stick-slip phenomena. To mitigate the adverse effects of Coulomb friction, friction-reduction techniques need to be implemented, such as the implementation of novel inerter configurations, low-friction materials, advanced surface treatment methods, and appropriate lubrication strategies. Wang et al. (2025) proposed a novel TMDI configuration in which the inerter is driven via high-strength ropes, as shown in Fig. 2(b). Through both experimental testing and numerical simulations on single-degree-of-freedom oscillators, they demonstrated the vibration control effectiveness of the proposed inerter-type TMD and further analyzed the detrimental effects of Coulomb damping on control performance.

Despite significant progress in the development of TMDI-based strategies for low-frequency VIV control of bridges, most existing studies remain numerical, with only limited experimental validation. In recent years, several research teams worldwide have developed various inerter devices in different configurations (Smith, 2020; Wagg, 2021). It is found that, in practice, inerter devices often exhibit excessive friction and pronounced nonlinearities, making it difficult to tune the system to its optimal parameters and thereby limiting their effectiveness in suppressing VIV. Among these issues, Coulomb friction is considered the most critical challenge (Wagg, 2021). For low-frequency bridge vibrations, the total force acting on the damper may be insufficient to overcome frictional resistance, causing the damper to "lock out" and

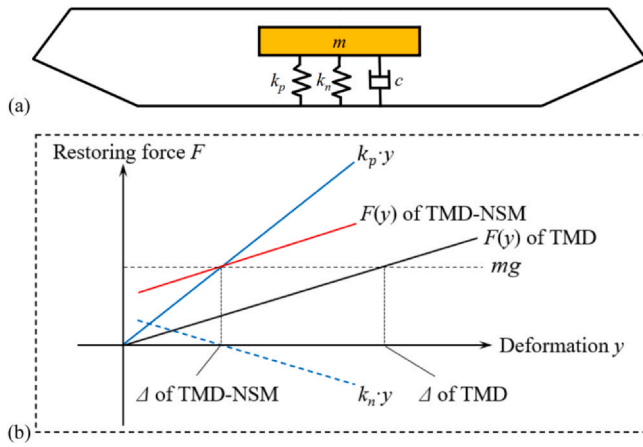


Fig. 3. Schematic diagram of (a) a TMD-NSM installed in a bridge deck and (b) its restoring force versus deformation.

remain inactive for much of the time, as observed in the low-frequency TMD case reported by Love and Haskett (2022). Therefore, there is a pressing need to develop low-friction inerter devices to improve the applicability of TMDIs in the low-frequency VIV control of bridges. Furthermore, in the studies reviewed above, the two terminals of the inerter element are typically positioned at the same location along the bridge's longitudinal direction. This configuration generally results in control effectiveness that is inferior to that of conventional TMDs. Notably, when the terminals are placed at different longitudinal locations, the control performance improves significantly as the terminal spacing increases. This aspect will be further discussed in Section 2.3.

For a vertical TMD, another effective approach to reduce the static deformation is to introduce a negative stiffness mechanism (NSM) in parallel with the positive stiffness element, resulting in a TMD-NSM as illustrated in Fig. 3, in which $k_p > 0$ represents the stiffness of the positive stiffness element and $k_n < 0$ represents the stiffness of the NSM. The NSM generates a restoring force in the same direction as the deformation, with a force-displacement curve (i.e., $k_n \cdot y$) exhibiting a negative slope. By proper system design, the positive stiffness element supports the self-weight of the mass block, while the positive stiffness element and NSM in parallel provide dynamic stiffness to the mass block. Consequently, the static deformation and natural frequency of the TMD-NSM are respectively calculated as:

$$\Delta = mg/k_p \quad (3)$$

$$f = \frac{1}{2\pi} \sqrt{(k_p + k_n)/m} \quad (4)$$

For the same mass m and tuning frequency f , the positive stiffness k_p of a TMD-NSM is significantly higher than the overall stiffness k of a traditional TMD, so that the static deformation of the TMD-NSM can be significantly reduced. One notable advantage of the TMD-NSM, compared to the TMDI, is that it maintains control effectiveness equivalent to that of the conventional TMD. The concept of NSM has been widely applied in low-frequency vibration isolation devices, commonly referred to as high-static-low-dynamic stiffness isolators or quasi-zero stiffness isolators. Relevant information can be found in recent review papers (Li et al., 2020, 2025).

Several studies have proposed improved TMD schemes that combine NSMs with TMDs. For instance, Kapasakalis et al. (2017) introduced the KDamper scheme, which grounds a conventional TMD by adding an NSM to enhance vibration control performance. However, for long-span bridges, TMDs are typically installed at locations such as the mid-span of the main girder where vibration amplitudes are large, making it impractical to ground the TMD via an NSM in such cases. Wang et al. (2022) pointed out that when structural degradation causes a decrease

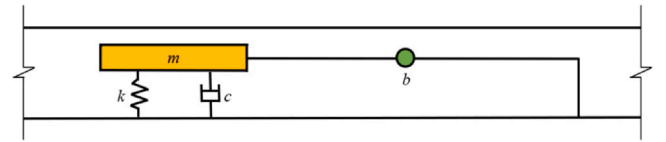


Fig. 4. Schematic diagram of a TMDI with inerter terminals connected to two distinct points along the bridge span.

in natural frequency and consequent detuning between the TMD and the structure, an NSM can be introduced to lower the TMD's natural frequency and re-tune it with the structure. These studies remain at the early stages of theoretical analysis and have yet to address low-frequency vertical vibration control of structures or the problem of excessive static deformation of TMD springs. Nevertheless, TMDs combined with NSMs offer a promising approach for low-frequency VIV control of long-span bridges.

2.3. TMD variants with lightweight design

When one terminal of the inerter in a TMDI is connected to the TMD mass block and the other to the ground (or another fixed point), the TMDI is equivalent to a conventional TMD with an augmented mass of $m + b$, where m is the mass of the TMDI mass block and b denotes the inertance. This mass amplification enables a lightweight TMD design by achieving control performance comparable to that of a large-mass TMD, while significantly reducing the required physical mass. In long-span bridge VIV control, TMDIs are typically installed near the mid-span of the main girder, where modal displacements are largest. In such cases, grounding one terminal of the inerter is often impractical. Alternatively, the inerter terminals can be connected to two distinct points along the bridge span, effectively utilizing the inerter's mass amplification effect, as illustrated in Fig. 4. Dai et al. (2019) showed that placing the two inerter terminals at different longitudinal positions on the bridge significantly enhances VIV suppression as the separation distance increases. Li et al. (2023) proposed to control bridge vibrations using the concept of a lever-type TMDI, in which two terminals of the inerter are connected to different points of bridge through a lever mechanism. Lei and Wang (2024) proposed a multiple TMDI system for mitigating VIV responses in parallel bridges, wherein two individual bridges are interconnected via two TMDIs. In this configuration, the mass block of each TMDI is mounted on one bridge, with one terminal of the inerter connected to the mass block and the other to the adjacent bridge. This inter-bridge coupling strategy demonstrated superior vibration suppression compared to separately installed TMDIs, offering a cost-effective passive control solution with reduced mass and inertance ratios.

Another promising strategy for achieving lightweight TMD design is the incorporation of NSMs. When one end of the NSM is connected to the TMD mass and the other to a fixed point, the vibration control performance of the TMD can be significantly improved. For long-span bridge applications, two ends of the NSM may be connected to different longitudinal locations on the bridge, thereby improving control efficiency—an approach analogous to that proposed by Su et al. (2023) for tall building vibration mitigation. This approach presents a viable solution for implementing lightweight TMDs in the VIV control of long-span bridges.

2.4. Nonlinear energy sink (NES)

The control effectiveness of TMDs is sensitive to detuning. When a bridge's natural frequency shifts due to temperature variation or structural degradation, the performance of TMDs can deteriorate significantly. Furthermore, a TMD is typically designed to control a single vibration mode, which limits its effectiveness under broadband excitations or in the control of multi-mode vibrations. In situations where

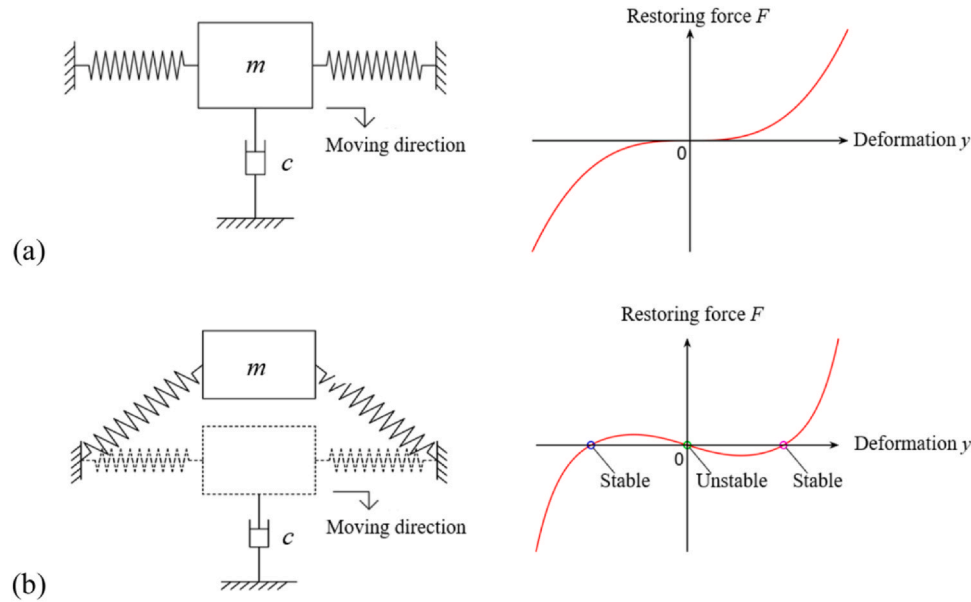


Fig. 5. Schematic diagrams of (a) a cubic-stiffness NES and (b) a bistable-stiffness NES.

VIVs involve multiple modes, several TMDs are often required, which increases both the complexity and the cost of the control system.

To overcome these limitations associated with frequency sensitivity and narrowband control effectiveness, a new type of passive vibration absorber known as the NES has been proposed (Vakakis et al., 2003). The core feature of the NES is its essential nonlinearity, which is typically achieved using a purely nonlinear stiffness element that lacks any linear component. This characteristic allows the NES to interact strongly with the primary structure without the need for precise frequency tuning. Theoretical studies have shown that NESs can achieve targeted energy transfer, where energy is irreversibly and unidirectionally transferred from the primary structure to the NESs over a broad frequency range. This mechanism gives NESs inherent advantages in coping with structural parameter uncertainties and broadband excitations, and makes them theoretically well suited for controlling multi-mode vibrations. The most extensively studied NES types are those with cubic stiffness and those with bistable stiffness. As illustrated in Fig. 5(a), cubic stiffness is typically achieved using a pair of horizontal springs that are initially undeformed. In contrast, as shown in Fig. 5(b), bistable stiffness is commonly realized using a pair of pre-compressed horizontal springs, which create two stable equilibrium positions and enable snap-through motion due to the presence of an intermediate unstable equilibrium. Comprehensive summaries of relevant research efforts can be found in recent review articles (Ding and Chen, 2020; Saeed et al., 2023).

In recent years, several studies have begun to investigate the application of NESs for controlling VIVs. Existing research has evaluated the effectiveness of NESs in suppressing VIV in cylindrical structures with one or two degrees of freedom (Mehmood et al., 2014; Chen et al., 2021; Zhang et al., 2022a). In the context of bridge VIV control, Zhang et al. (2022b) compared the performance of a TMD and a cubic-stiffness NES for a continuous beam structure. Their results showed that while the TMD provides better control under optimal tuning, the NES exhibits greater robustness to frequency variations and offers a certain level of multi-mode control even with a single unit. Wu et al. (2025) also conducted a comparative study on the VIV control performance of a cubic-stiffness NES and several TMD variants, using a numerical example of a bridge deck sectional model. Farhangdoust et al. (2020) investigated the VIV control in a bridge using a bistable NES and proposed an optimization algorithm for tuning the parameters of the

bistable NES. Yu et al. (2022) investigated the performance of an NES inerter (NESI) for bridge VIV suppression and demonstrated its potential for multi-mode control. It was found that the performance of the NESI strongly depends on the locations of the inerter terminals, and the NESI outperforms the NES as long as the difference between the modal displacements at the two terminals exceeds a certain (usually small) threshold. Xie et al. (2024) proposed a strategy using a bistable NESI for multi-mode VIV control in bridge structures, demonstrating lower stiffness requirements and superior control performance compared to the cubic-stiffness NESI studied by Yu et al. (2022).

Despite these advances, NES-based VIV control for bridges remains at an early stage, with several challenges still to be addressed. One major issue is that NES performance is sensitive to the initial energy of the system. This limitation can be partially alleviated by optimizing the nonlinear stiffness characteristics of the NES or by adopting multi-degree-of-freedom NES configurations (Ding and Chen, 2020; Saeed et al., 2023). Furthermore, existing research is primarily numerical and relies on empirical vortex-induced force models, with a lack of reliable experimental validation. Although several physical NES models have been developed and tested in laboratories, most are designed for horizontal vibration control. When applied to vertical vibration control, the self-weight of the NES mass acts on the nonlinear spring and causes a static displacement. Unlike linear TMDs, this static displacement alters the dynamic characteristics of the NES, making the effect of self-weight on vertical vibration control performance significant, especially for low-frequency structures (Zhang et al., 2025).

To address this issue, Chen et al. (2023) proposed a vertical NES design using a disc spring to support the self-weight of the NES mass and to generate the desired nonlinear stiffness. However, the maximum stroke of this NES was limited by the deformation capacity of the disc spring. Li and Ding (2024) introduced a vertical-track NES exhibiting cubic stiffness at its static equilibrium position and demonstrated its performance with a physical model featuring a maximum stroke of 3 mm. Liao et al. (2024) developed a cam-based vertical NES with bistable stiffness and tested its vibration control performance using a physical model. However, its maximum stroke was also limited to only a few millimeters. Currently, no vertical NES device is fully suitable for controlling VIVs in long-span bridge applications.

2.5. Viscous damper

The energy dissipation mechanism of viscous dampers relies on substantial relative motion between their two ends. To effectively apply

viscous dampers for controlling VIVs of bridge decks, adequate relative displacement must be ensured. Several studies have explored the potential of viscous dampers for bridge VIV control. Møller et al. (2019) proposed a damping system comprising four devices installed between the pylons and the main suspension cables. Each device includes a viscous damper and a spring arranged in parallel and is connected to the pylon and main cable via a pre-tensioned link. Chen et al. (2022) and Liu et al. (2023) observed that the ends of the bridge deck experience significant rotational motion during vertical vibrations in suspension bridges. Based on this observation, they proposed an outrigger system incorporating viscous dampers between the deck ends and the towers or piers to suppress VIVs. Peng et al. (2025b) proposed a method for mitigating wind-induced vibrations in long-span bridges using a compound damped cable system, in which a viscous damper and a spring are arranged in parallel. One end of the system is anchored to the ground, while the other end is connected to the bridge deck via cables. Cao et al. (2023, 2025) introduced a control scheme in which viscous dampers are placed between the main girder and the transverse beams of the towers. This configuration is particularly suited to suspension bridges with separated deck-tower arrangements; for integrated deck-tower systems, additional bracket structures are required for damper installation. Wind tunnel tests and numerical simulations have demonstrated the effectiveness of this approach, which has been successfully applied to the VIV control of the Yingwuzhou Yangtze River Bridge.

3. Active device

Unlike passive devices that rely on fixed mechanical properties, active devices utilize external energy and real-time feedback to actively mitigate structural vibrations. A typical active vibration control system comprises sensors, actuators, and a control algorithm that modulates the applied forces based on the measured structural response. Common actuators include hydraulic and electromagnetic types. This adaptive force modulation enables active devices to achieve superior performance, particularly in complex or time-varying environments.

Numerous studies have demonstrated the feasibility and effectiveness of using active control devices for vibration mitigation in civil engineering structures. Among them, the most extensively studied are active TMDs and active mass dampers (AMDs), as illustrated in Fig. 6. A comprehensive review of recent advances in the development and application of active TMDs and AMDs for building-like structures is provided by Koutsoloukas et al. (2022). Additionally, a critical review on recent developments in control algorithms can be found in Wani et al. (2022). Active control devices have also been employed to mitigate VIVs of bridge towers, e.g., the Osman Gazi Suspension Bridge tower during both construction and in-service phases (Inoue et al., 2025). In the context of VIV control for bridge decks, Battista and Pfeil (2000) compared numerically the performance of AMDs and conventional TMDs on the Rio-Niterói Bridge, concluding that AMDs offered superior vibration suppression. Yu (2013) numerically compared the VIV control performance of a passive TMD, an active TMD, and an AMD using a bridge deck case study. The results demonstrated that both the active TMD and AMD are effective solutions in scenarios involving multimode VIVs or where static deformations must be spatially constrained. Dai et al. (2022a) examined VIV control of a continuous beam bridge using an active TMD and proposed a parameter-independent control

algorithm to enhance robustness against uncertainties in the bridge's natural frequency.

Compared to passive control devices, active control devices offer several significant advantages: (1) they can continuously adjust the control force in real time based on feedback, enabling effective performance under varying structural and environmental conditions; (2) they provide superior vibration mitigation, especially in scenarios where passive devices may be less effective or improperly tuned; (3) unlike passive devices, which are typically limited to a narrow frequency band, active devices can target a broader range of excitation frequencies; (4) active devices can achieve comparable or better vibration control with significantly smaller auxiliary masses; and (5) active devices can be programmed to simultaneously suppress multiple vibration modes, thereby enhancing multimodal control capabilities relative to passive counterparts.

To the best of the authors' knowledge, active devices have not yet been implemented for VIV control of bridge decks despite their clear advantages. This is likely due to challenges including high power requirements, the need for robust control systems, and the requirement for long-term reliability in harsh outdoor environments. Additionally, the complexity and cost of sensor-actuator networks and real-time controllers present significant barriers to large-scale implementation on long-span bridges. Spencer and Nagarajaiah (2003) also noted that the civil engineering sector and construction industry tend to be conservative and cautious about adopting new technologies. The lack of standardized, verified, and widely accepted design, analysis, and testing protocols further hinders practical application. Nevertheless, the continuous advancement of active control technology holds potential for enabling adaptive, lightweight, and intelligent control in future smart infrastructure.

4. Semi-active device

Semi-active control devices have emerged as a promising middle ground between passive and active vibration control systems, combining the reliability and low energy consumption of passive devices with the adaptability of active systems. Semi-active devices can adjust their damping or stiffness properties in real time without requiring large external power inputs. Typical semi-active devices include magnetorheological (MR) dampers, electrorheological (ER) dampers, variable friction dampers, and variable stiffness devices. Among these, MR dampers have been the most extensively studied for civil infrastructure due to their rapid response time, wide controllable range, and relatively low power consumption. By applying a controlled magnetic field, the rheological properties of MR fluids can be modulated, enabling variable damping forces that can be tuned to the bridge's dynamic response. Fig. 7 illustrates a schematic of a semi-active TMD, in which both stiffness and damping coefficients are continuously tunable based on real-time feedback.

Compared with fully active devices, semi-active systems offer several advantages: (1) they require only small amounts of power for adjusting device properties, often supplied by low-voltage control circuits; (2) in the event of power loss, semi-active devices default to passive damping modes, maintaining some level of vibration mitigation; (3) semi-active devices generally have simpler mechanical structures and control requirements, reducing vulnerability to failures and

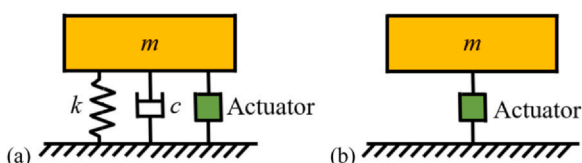


Fig. 6. Schematic diagrams of (a) an active TMD and (b) an AMD.

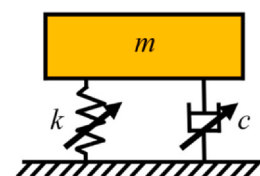


Fig. 7. Schematic diagram of a semi-active TMD with tunable stiffness and damping coefficients.

environmental degradation; (4) lower complexity and energy consumption translate to reduced installation and operational costs, making them more feasible for large-scale bridge applications.

Numerous experimental and numerical studies have demonstrated the effectiveness of semi-active devices in suppressing vibrations on civil engineering structures (Gkatzogias and Kappos, 2016). A notable application of semi-active VIV control is the MR semi-active TMD for controlling the VIVs of the Volgograd Bridge. The main feature of the MR semi-active TMD concept is that the real-time controlled MR damper emulates a controllable stiffness force and a controllable damping force. The controllable stiffness force augments or diminishes the stiffness of the passive springs and thereby tunes the MR semi-active TMD frequency to the actual vibration frequency of the bridge (Weber et al., 2011; Weber and Maślanka, 2012). The controllable damping force generates frequency-dependent energy dissipation. Details of the MR semi-active TMD system are reported in Weber et al. (2013). The system comprises 12 semi-active TMDs, designed to target three vibration modes at 0.45, 0.57, and 0.68 Hz, respectively. For each mode, the passive mass-spring components of four devices are tuned to its nominal frequency. However, all 12 devices are controlled in real-time to the actual vibration frequency; hence all 12 semi-active TMDs mitigate the actual mode of vibration. As a direct result, the MR-STMD concept generates approximately the same vibration reduction in the bridge as if passive TMDs with approximately twice as much mass were installed.

More recently, Dai et al. (2021b) proposed a control strategy that considers the effect of VIV characteristics on the control force determination. A control strategy considering VIV characteristics is proposed for realizing the real-time tuning and mass stroke limitation simultaneously. Liu et al. (2025) introduced a method to control the VIV of a bridge deck using a variable-spacing ball screw eddy current damper, in which the damping coefficient is continuously tunable by adjusting the spacing between the surface of the permanent magnet and the conductor plate.

5. Recommendations for future direction

Building upon the review of energy dissipative devices for VIV control in bridges, this section outlines key recommendations and future development directions to improve the performance, feasibility, and adoption of these devices in practical engineering applications. These recommendations aim to address current technical limitations, bridge gaps between research and implementation, and support the evolution of smarter, more resilient infrastructure.

5.1. Development of TMD prototypes for low-frequency VIV control

Many long-span bridges, particularly suspension types, have natural frequencies in the ultra-low range (e.g., below 0.2 Hz), where traditional TMDs become inapplicable due to excessive spring deformation. Consequently, there is a pressing need to develop novel TMD prototypes specifically tailored for this frequency regime. Future research should focus on designing effective, reliable, and structurally compatible TMD systems capable of operating under such demanding conditions. Promising directions include the incorporation of advanced mechanical configurations, such as inerter-based mechanisms, NSMs, and high-static-low-dynamic-stiffness systems.

5.2. Lightweight and compact device design

The integration of energy dissipative devices into long-span bridges presents several practical challenges, particularly concerning spatial constraints, added dead load, and long-term maintenance requirements. Excessive device weight not only imposes additional demands on structural components such as girders and towers but also complicates fabrication, transportation, and on-site installation. Therefore, future

research and development should prioritize the design of lightweight and compact damping devices that offer high energy dissipation efficiency without imposing undue structural penalties.

5.3. Advancement of multimode passive control devices

VIVs in bridge decks may be dominated by different structural modes with changing wind velocity. This modal variability poses a significant challenge to conventional passive devices like TMDs, which are typically designed to target a single mode. Future research should focus on the development of advanced passive devices capable of providing effective and robust multimode vibration mitigation. One promising direction involves the design and prototyping of devices incorporating nonlinear stiffness mechanisms, such as NESs. These devices are capable of interacting with a broad spectrum of vibration frequencies, thereby enabling effective multimode control with a reduced number of units.

5.4. Enhancement of active and semi-active control systems

Active and semi-active vibration control systems have demonstrated superior flexibility and performance compared to passive systems. However, challenges remain in terms of control stability, energy consumption, cost, and long-term reliability. In particular, these systems often rely on complex algorithms and real-time feedback loops that can be sensitive to modeling inaccuracies, sensor noise, and time delays. This may lead to instability or degraded performance under varying environmental conditions or unexpected disturbances. Additionally, active systems generally involve high energy consumption, which raises concerns over operational costs and sustainability. Long-term reliability is another critical issue, as the performance of electronic components can deteriorate over time due to fatigue, environmental degradation (e.g., temperature variations, corrosion), and mechanical wear. To enhance the practical value of active and semi-active control systems, future efforts should focus on improving control algorithms, particularly robust, adaptive, or learning-based approaches that can maintain performance in the presence of environmental uncertainties and nonlinear structural behavior. At the hardware level, advances in compact actuators, fault-tolerant electronics, and energy-efficient components will be essential. Semi-active systems, particularly those based on MR fluids, remain a promising compromise, offering adaptability with minimal power consumption and fail-safe passive behavior in power outages.

5.5. Exploration of self-powered and energy-harvesting control devices

One of the most significant barriers to widespread implementation of active devices in bridge applications is their reliance on continuous external power, which not only increases operational complexity but may also introduce instability due to external energy input. Future developments should explore self-powered and energy-harvesting control devices that can operate autonomously. For instance, piezoelectric, triboelectric, or electromagnetic transducers could be used to harvest energy from bridge VIVs as well as the vibration of the dampers. By integrating such harvesters with compact energy storage components (e.g., supercapacitors or rechargeable batteries), it may become feasible to realize fully autonomous, self-sustaining systems.

5.6. Multi-hazard vibration control strategies

Modern bridge structures are increasingly exposed to multiple hazards, such as wind, wave, and seismic events, that may occur simultaneously or sequentially. Traditional vibration mitigation strategies are often designed for a single dominant hazard, which may limit performance under real-world loading conditions. A shift toward integrated multi-hazard control is therefore needed. Future research

should investigate hybrid control systems that can recognize and respond to different hazard types, adjust control modes accordingly, and maintain structural safety and serviceability under compound effects.

5.7. Integration of smart control and digital twin technologies

The integration of energy dissipative devices with smart technologies such as digital twins and AI-based control systems represents a promising direction for VIV control in long-span bridges. Digital twin frameworks, continuously updated with real-time sensor data, can serve as virtual replicas of the physical structure, enabling enhanced structural health monitoring, predictive maintenance, and dynamic optimization of control strategies. AI and machine learning algorithms can further improve control adaptability and robustness by autonomously adjusting to changing environmental and structural conditions. Despite these promising developments, several barriers hinder the widespread implementation of energy dissipative devices integrated with smart control in long-span bridges. These include challenges related to data fusion from heterogeneous sensors, the complexity of cyber-physical system integration, the stability of control algorithms, high installation and maintenance costs, and ensuring system reliability under harsh environmental conditions. Future research should focus on overcoming these technical and practical obstacles to realize the full potential of intelligent vibration mitigation systems, thereby enhancing bridge safety, longevity, and operational efficiency.

5.8. Standardized design and application guidelines

Despite extensive academic research, the practical deployment of advanced energy dissipative devices for bridge VIV control remains limited. A major obstacle is the lack of standardized procedures for design, testing, validation, and implementation. Future efforts should aim to establish comprehensive design guidelines and technical standards tailored to bridge structures. These should cover device selection criteria, expected performance ranges, environmental durability requirements, construction integration strategies, and long-term monitoring protocols. Standardization would not only facilitate consistent performance evaluation across different projects but also promote confidence among engineers, contractors, and regulatory bodies, accelerating industry adoption.

6. Conclusion

This study provides a comprehensive and up-to-date review of energy dissipative devices employed for controlling VIVs in bridge decks, encompassing passive, active, and semi-active control systems. It synthesizes key findings and innovative control strategies proposed within the VIV control community, critically evaluating the inherent advantages and limitations of each device type and identifying essential research gaps for future investigation.

Several documented practical implementations of these devices for VIV control in existing long-span bridges are reviewed, revealing that TMDs remain the most widely adopted solution, with successful deployment in ten bridge projects. In contrast, semi-active TMDs and viscous dampers have been applied in only one bridge each, while no documented cases of active device implementation for VIV control in real bridge decks exist to date.

While TMDs have proven effective, optimizing their parameters, particularly under nonlinear vortex-induced forces and closely spaced modal frequencies, continues to be a prominent research focus. Recent advances emphasize multi-objective optimization approaches that balance various performance criteria simultaneously to enhance control efficacy.

Despite their advantages, traditional TMDs face challenges such as limited frequency bandwidth, unsuitability for ultra-low-frequency modes, and reduced effectiveness in multimodal vibration scenarios.

These issues have stimulated the development of advanced passive devices for low-frequency VIV control, lightweight and compact device configurations, and multimode vibration mitigation. Alternative damping devices such as viscous dampers also show promise. However, the majority of existing studies are numerical or based on small-scale laboratory experiments, highlighting the pressing need for prototype development and full-scale testing to validate these approaches and facilitate practical adoption.

Active and semi-active vibration control devices offer greater adaptability and enhanced performance compared to passive systems. While both active and semi-active technologies have demonstrated superior vibration mitigation capabilities in research and small-scale experiments, their widespread implementation in long-span bridges remains limited due to cost, complexity, and the need for robust, adaptive control algorithms.

Future research should focus on developing lightweight, compact, and multimode-capable prototype devices, improving control strategies with adaptive and learning-based algorithms, and advancing self-powered or energy-harvesting systems to overcome power supply constraints. Integration of smart control technologies and digital twins promises transformative potential for real-time monitoring, predictive maintenance, and optimized vibration mitigation. Finally, establishing standardized design guidelines and full-scale validation protocols will be critical to bridging the gap between laboratory research and practical engineering deployment, ultimately enhancing the resilience and longevity of long-span bridges subjected to VIVs.

CRediT authorship contribution statement

Mingjie Zhang: Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Haiyan Yu:** Writing – original draft, Visualization, Resources, Investigation. **Zhanbiao Zhang:** Writing – original draft, Visualization. **Fuyou Xu:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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